```
pip install pypdf matplotlib wordcloud
```

```
Collecting pypdf
  Downloading pypdf-6.6.2-py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.
Requirement already satisfied: wordcloud in /usr/local/lib/python3.12/dist-packages (1.9
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packag
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Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (frc
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from
Downloading pypdf-6.6.2-py3-none-any.whl (329 kB)
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Installing collected packages: pypdf
Successfully installed pypdf-6.6.2
```
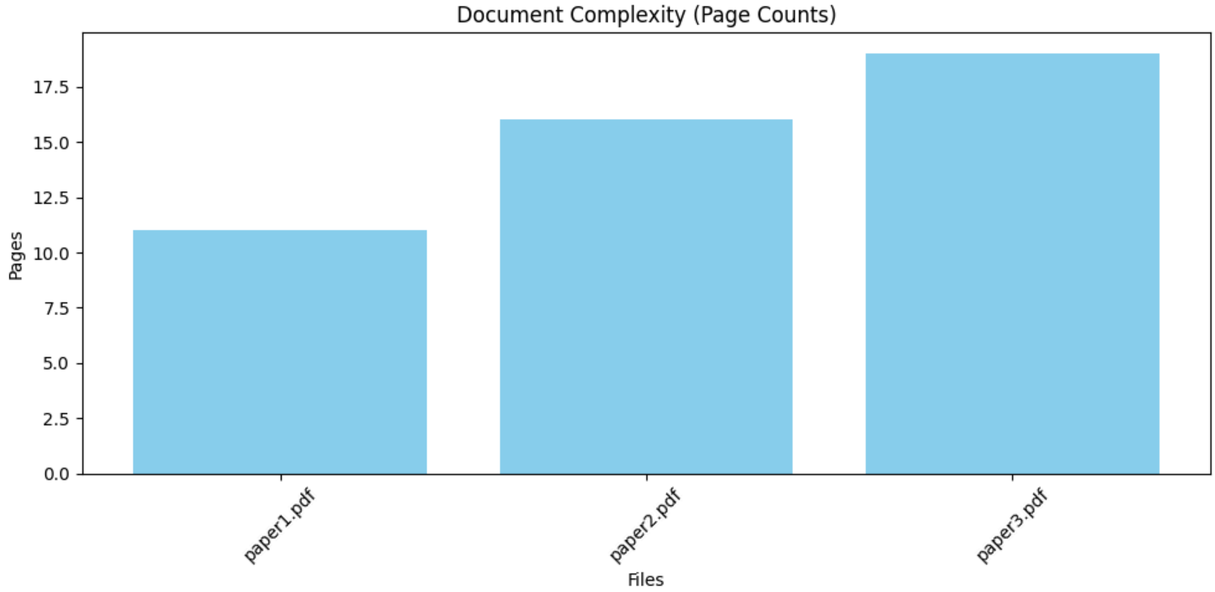
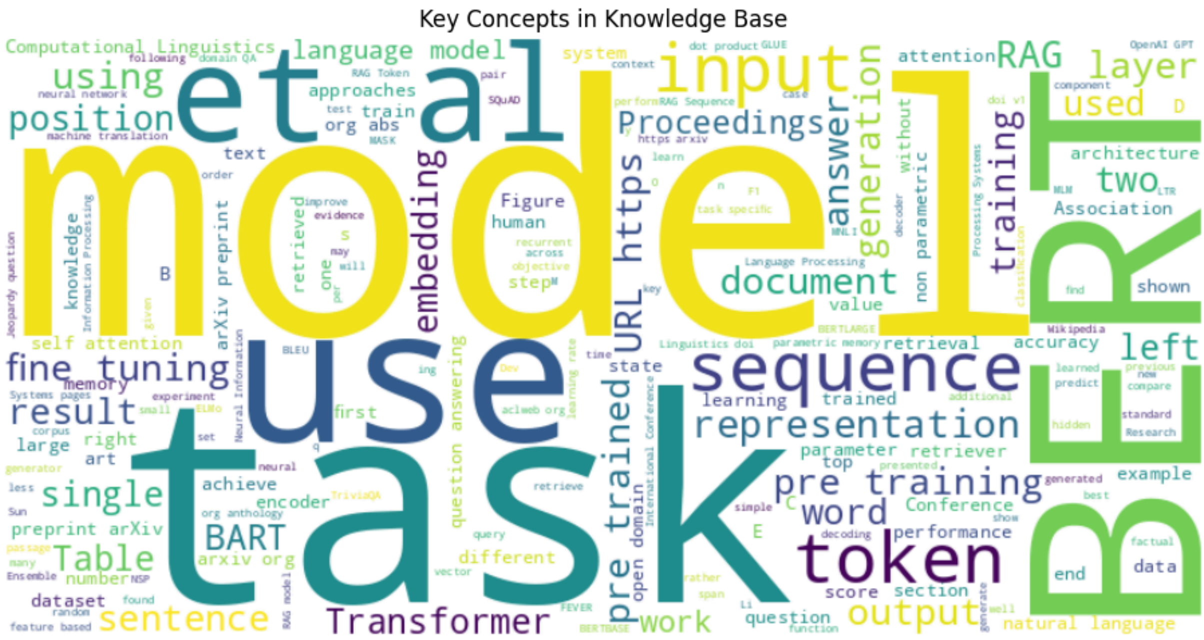

```
mport os
rom pypdf import PdfReader
mport matplotlib.pyplot as plt
rom wordcloud import WordCloud

 1. LOAD DATA
ata_folder = '/content/Data'  # Point to your data folder
ll_text = ""
age_counts = []

rint(f"Loading PDFs from {data_folder}...")
or filename in os.listdir(data_folder):
   if filename.endswith('.pdf'):
       reader = PdfReader(os.path.join(data_folder, filename))
       num_pages = len(reader.pages)
       page_counts.append((filename, num_pages))

       # Extract text for WordCloud
       for page in reader.pages:
           all_text += page.extract_text() + " "

 2. GENERATE GRAPH 1: Document Lengths
ames = [x[0] for x in page_counts]
ounts = [x[1] for x in page_counts]

lt.figure(figsize=(10, 5))
lt.bar(names, counts, color='skyblue')
lt.title('Document Complexity (Page Counts)')
lt.xlabel('Files')
lt.ylabel('Pages')
lt.xticks(rotation=45)
lt.tight_layout()
lt.savefig('eda_page_counts.png') # Save to show in report
lt.show()

 3. GENERATE GRAPH 2: Knowledge Word Cloud
ordcloud = WordCloud(width=800, height=400, background_color='white').generate(all_text)

lt.figure(figsize=(10, 5))
lt.imshow(wordcloud, interpolation='bilinear')
lt.axis('off')
lt.title('Key Concepts in Knowledge Base')
lt.tight_layout()
```

```
lt.savefig('eda_wordcloud.png') # Save to show in report
lt.show()

rint("EDA Completed. Images saved.")
```

```
Loading PDFs from /content/Data...
```



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EDA Completed. Images saved.
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